

LOCALIZED BOST–CONNES SYSTEMS I: FACTOR TYPE AND THE SPECTRUM OF TRANSITIONS

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ABSTRACT. The obstruction group $\Xi_\beta(K, S)$ of [13] classifies the KMS_β simplex. We show that it is the $s = 0$ slice of a single larger object, the *extended obstruction group*

$$\Xi_\beta^{\text{ext}}(K, S) = \left\{ (\chi, s) \in \widehat{G_S} \times \mathbb{R} : \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}}) N\mathfrak{p}^{i\beta s}|^2 < \infty \right\},$$

whose slices $s \neq 0$ control the Murray–von Neumann–Connes type of the associated factors from above: the essential values of the Radon–Nikodym cocycle annihilate $\Sigma_\beta = \text{pr}_{\mathbb{R}} \Xi_\beta^{\text{ext}}$, so $\Sigma_\beta \neq \{0\}$ excludes type III_1 , a dense Σ_β forces III_0 , and $\Sigma_\beta = c\mathbb{Z}$ forces III_{λ^k} with $\lambda = e^{-2\pi/c}$ or III_0 ; moreover Σ_β lies in the point spectrum of the flow of weights. Matching lower bounds come from a direct construction of essential values in the manner of Araki and Woods. The phase classification and the type classification are thus two coordinate slices of one Dirichlet-series condition. For $S = P_K$ and $\beta = 1$ both reduce to the non-vanishing of Hecke L -functions on $\text{Re} = 1$: the $s = 0$ slice is $L(1, \chi) \neq 0$ and the $s \neq 0$ slices are $L(1 - i\tau, \chi) \neq 0$.

Two consequences are perhaps unexpected. First, the natural sufficient condition for type III_1 — that two primes have irrationally related log-norms — is false: two primes contribute a finite amount to a divergent series. We exhibit $S \subset P_{\mathbb{Q}}$ with $\zeta_S(1) = \infty$, a *unique* KMS_1 state, and factors of pure type $\text{III}_{e^{-1}}$; uniqueness of the KMS state and type III_1 are logically independent. Second, for infinite S the log-norms are never pairwise commensurable, so any failure of III_1 comes from *approximate* commensurability along a set of divergent β -weight.

In the horizontal direction we determine which chains $\{\Xi_\beta\}_{\beta>0}$ occur. For every sequence $(\beta_n) \subset (0, 1)$ there is S with $\beta_c(S) = 1$, dense parameter subgroup, and $\Xi_\beta = \bigoplus_{n: \beta_n < \beta} \mathbb{Z}/2\mathbb{Z}$; taking (β_n) dense in $(0, 1)$ gives countably many dense transitions, and taking it dense in a Cantor set $\mathcal{C}$ makes the non-local-constancy locus exactly $\mathcal{C}$. The continuous devil’s staircase, however, is unattainable for a structural reason: $\widehat{G_S}$ is countable for every S , so the jump set is countable and every order parameter is a pure jump function. It is approachable in the uniform norm and not attained.

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1. INTRODUCTION: ONE GROUP, TWO DIRECTIONS

We use the notation of [13]: $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ with $\sigma_t(\mu_a) = N a^{it} \mu_a$, symmetry group G_S , parameter homomorphism $\sigma : I_S \rightarrow G_S$, and

$$(1.1) \quad \Xi_\beta(K, S) = \left\{ \chi \in \widehat{G_S} : \sum_{\mathfrak{p} \in S} N \mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2 < \infty \right\},$$

the KMS_β simplex being $\text{Prob}(G_S/\Xi_\beta^\perp)$. Write $\mathcal{R}_S$ for the tail equivalence relation on $V = \mathbb{Z}_{\geq 0}^S$ and π_β for the forced product measure [13, Lem. 2.10].

The series. The foundational paper is [13]. The present paper is the first of a series continuing it; the later papers, in preparation, treat the ideal structure and boundary quotient, the invariants that see Ξ_β and those that do not, the metric geometry, and the free variant. Nothing here depends on them.

The phase diagram of $\mathcal{A}_{K,S}$ has two directions. Horizontally one asks which β are transition points, i.e. how the chain $\{\Xi_\beta\}$ grows; vertically one asks what the factors at a given β are. Definition 1.1 shows the two questions are slices of one object, and the rest of the paper answers both — the second one partly: the object bounds the type from above, and the bound is matched by direct constructions in the cases treated.

Definition 1.1. The *extended obstruction group* is

$$(1.2) \quad \Xi_\beta^{\text{ext}}(K, S) = \left\{ (\chi, s) \in \widehat{G_S} \times \mathbb{R} : \sum_{\mathfrak{p} \in S} N \mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}}) N \mathfrak{p}^{i\beta s}|^2 < \infty \right\},$$

a subgroup of $\widehat{G_S} \times \mathbb{R}$, and $\Sigma_\beta := \text{pr}_{\mathbb{R}}(\Xi_\beta^{\text{ext}}) \leq \mathbb{R}$.

That Ξ_β^{ext} is a subgroup follows as in [13, Lem. 3.5] from $|1 - z_1 z_2| \leq |1 - z_1| + |1 - z_2|$ for $|z_i| = 1$; and $\Xi_\beta = \Xi_\beta^{\text{ext}} \cap (\widehat{G_S} \times \{0\})$.

2. THE TYPE CRITERION

Let $\beta > 0$ with $\zeta_{K,S}(\beta) = \infty$, let φ be an extremal KMS_β state and $M = \pi_\varphi(\mathcal{A}_{K,S})''$. By [13, Prop. 2.8 and Lem. 2.10], φ is given by a probability measure ν on $Y = V \times G_S$ whose V -marginal is π_β , and M is the von Neumann algebra of the orbit equivalence relation $\mathcal{R}$ on (Y, ν) generated by the maps $(v, h) \mapsto (v + e_{\mathfrak{p}}, \sigma_{\mathfrak{p}} h)$, $\mathfrak{p} \in S$. The Radon–Nikodym cocycle of $\mathcal{R}$ is

$$(2.1) \quad \rho(y, \mathfrak{a} \mathfrak{b}^{-1} \cdot y) = \log \frac{d\nu \circ (\mathfrak{a} \mathfrak{b}^{-1})}{d\nu}(y) = -\beta \log \frac{N \mathfrak{a}}{N \mathfrak{b}},$$

and we write $E \subseteq \mathbb{R}$ for its group of finite essential values in the sense of [12]: $u \in E$ iff for every $B \subseteq Y$ with $\nu(B) > 0$ and every $\varepsilon > 0$ there is a partial transformation T of $\mathcal{R}$ with $\nu(B \cap T^{-1}B) > 0$ and $|\rho(y, Ty) - u| < \varepsilon$ on its domain. E is a closed subgroup of $\mathbb{R}$ [12], M is a factor of type III [13, Prop. 6.1], and by Krieger’s theorem [7] the Connes invariant is

$$(2.2) \quad S(M) = \{0\} \cup \exp(E),$$

so that M is of type III_1 , III_λ or III_0 according as $E = \mathbb{R}$, $E = (\log \lambda)\mathbb{Z}$ with $0 < \lambda < 1$, or $E = \{0\}$ [4]. Everything in this section is a computation of E .

Lemma 2.1 (Coboundary dictionary). *Let $(\chi, s) \in \widehat{G_S} \times \mathbb{R}$ and $c_{\mathfrak{p}} = \chi(\sigma_{\mathfrak{p}}) N \mathfrak{p}^{i\beta s}$. The following are equivalent:*

- (1) *there is a measurable $g : V \rightarrow \mathbb{T}$ with $g(v + e_{\mathfrak{p}}) = c_{\mathfrak{p}} g(v)$ for π_β -a.e. v , for every $\mathfrak{p} \in S$;*
- (2) $(\chi, s) \in \Xi_\beta^{\text{ext}}$.

Proof. Propositions 3.3 and 3.4 of [13] prove exactly this equivalence for $c_p = \overline{\chi(\sigma_p)}$, and their proofs use only that $|c_p| = 1$, that the coordinates v_p are independent under π_β with $\pi_\beta(v_p = k) = (1 - t_p)t_p^k$, $t_p = Np^{-\beta}$, and the estimate $1 - |m_p|^2 \asymp t_p|1 - c_p|^2$ of [13, Lem. 3.2] for $m_p = (1 - t_p)/(1 - c_p t_p)$, which holds for every unimodular c_p . (Conjugating g replaces c_p by $\overline{c_p}$, and $|1 - c_p| = |1 - \overline{c_p}|$.) $\square$

Proposition 2.2 (The extended group bounds the type). *Put $\Sigma_\beta^\vee = \{u \in \mathbb{R} : su \in 2\pi\mathbb{Z} \text{ for all } s \in \Sigma_\beta\}$. Then $E \subseteq \Sigma_\beta^\vee$. Consequently:*

- (1) *if $\Sigma_\beta \neq \{0\}$, then M is not of type III_1 ;*
- (2) *if Σ_β is dense in $\mathbb{R}$, then M is of type III_0 ;*
- (3) *if $\Sigma_\beta = c\mathbb{Z}$ with $c > 0$, then M is of type III_{λ^k} for some integer $k \geq 1$, where $\lambda = e^{-2\pi/c}$, or of type III_0 .*

Proof. Let $(\chi, s) \in \Xi_\beta^{\text{ext}}$, let g be as in Lemma 2.1, and put $G(v, h) = g(v)\overline{\chi(h)}$, a unimodular measurable function on Y . For $\mathbf{a} \in J_S$,

$$G(v + \mathbf{a}, \sigma_{\mathbf{a}}h) = \chi(\sigma_{\mathbf{a}})N\mathbf{a}^{i\beta s}g(v)\overline{\chi(\sigma_{\mathbf{a}})}\overline{\chi(h)} = N\mathbf{a}^{i\beta s}G(v, h),$$

so by (2.1) the $\mathbb{T}$ -valued cocycle $e^{-is\rho}$ on $\mathcal{R}$ is the coboundary $e^{-is\rho(y, y')} = G(y')/G(y)$. A coboundary has no essential value other than 1: if $|G - \gamma| < \delta$ on a set B_δ of positive measure, every pair $y, y' \in B_\delta$ has $|G(y')/G(y) - 1| < 2\delta$. On the other hand, if $u \in E$ then e^{-isu} is an essential value of $e^{-is\rho}$, since $|e^{-is\rho} - e^{-isu}| \leq |s||\rho - u|$. Hence $e^{-isu} = 1$ for all $u \in E$ and all $s \in \Sigma_\beta$, i.e. $E \subseteq \Sigma_\beta^\vee$.

(1) If $\Sigma_\beta \neq \{0\}$ then $\Sigma_\beta^\vee \neq \mathbb{R}$, so $E \neq \mathbb{R}$. (2) If Σ_β is dense then $\Sigma_\beta^\vee = \{0\}$, so $E = \{0\}$ and $S(M) = \{0, 1\}$. (3) If $\Sigma_\beta = c\mathbb{Z}$ then $\Sigma_\beta^\vee = \frac{2\pi}{c}\mathbb{Z}$, and the closed subgroups of $\frac{2\pi}{c}\mathbb{Z}$ are $\{0\}$ and $\frac{2\pi k}{c}\mathbb{Z}$, $k \geq 1$; by (2.2) these give $S(M) = \{0, 1\}$ and $S(M) = \{0\} \cup (e^{-2\pi k/c})^\mathbb{Z}$ respectively. $\square$

Remark 2.3 (Σ_β and the flow of weights). The proof shows more. With G as above, $\Psi(y, u) = G(y)e^{isu}$ is invariant under the Maharam extension $(y, u) \mapsto (y', u + \rho(y, y'))$ of $\mathcal{R}$ [10] and satisfies $\Psi(y, u + t) = e^{ist}\Psi(y, u)$; so every $s \in \Sigma_\beta$ is an eigenvalue of the flow of weights of M , i.e. Σ_β is contained in the point spectrum of that flow, which by Connes–Takesaki [5] is Connes’ invariant $T(M)$. We shall not need the converse inclusion and do not prove it; it holds when the flow of weights has pure point spectrum, and in that case Σ_β alone determines the type. Whether the flow of weights of M always has pure point spectrum is not known to us, and this is why Proposition 2.2 is only an upper bound. Lower bounds are obtained by exhibiting essential values directly, which is the purpose of the next lemma.

Lemma 2.4 (Asymptotic pairs). *Suppose $\Xi_\beta = \{1\}$, so that $\nu = \pi_\beta \otimes \text{Haar}$ [13, Thm. 3.6]. Let $u \in \mathbb{R}$ and let $(\mathbf{a}_j, \mathbf{b}_j)_{j \geq 1}$ be pairs of primes of S , all the primes $\mathbf{a}_j, \mathbf{b}_j$ distinct, such that*

$$\beta \log \frac{N\mathbf{a}_j}{N\mathbf{b}_j} \rightarrow u, \quad g_j := \sigma_{\mathbf{b}_j}\sigma_{\mathbf{a}_j}^{-1} \rightarrow 1 \text{ in } G_S, \quad \sum_j N\mathbf{a}_j^{-\beta} = \infty.$$

Then $u \in E$.

Proof. Write $t_p = Np^{-\beta}$ and, for each j , $P_j = \{v_{\mathbf{a}_j} = 1, v_{\mathbf{b}_j} = 0\}$, $Q_j = \{v_{\mathbf{a}_j} = 0, v_{\mathbf{b}_j} = 1\}$, events of π_β -probability $p_j = (1 - t_{\mathbf{a}_j})(1 - t_{\mathbf{b}_j})t_{\mathbf{a}_j}$ and $q_j = (1 - t_{\mathbf{a}_j})(1 - t_{\mathbf{b}_j})t_{\mathbf{b}_j}$. The events $P_j \cup Q_j$ are independent and $\sum_j p_j = \infty$; since $t_{\mathbf{b}_j}/t_{\mathbf{a}_j} \rightarrow e^u$, there is $\theta > 0$ with $p_j \geq \theta(p_j + q_j)$ for all large j . Fix $\varepsilon > 0$ and j_0 , and let D be the set of (v, h) such that some $j \geq j_0$ has $v \in P_j \cup Q_j$ and the least such j , denoted $j(v)$, has $v \in P_j$. Define

$$T(v, h) = (v - e_{\mathbf{a}_j} + e_{\mathbf{b}_j}, g_j h), \quad j = j(v),$$

on D . The move changes only the coordinates $\mathfrak{a}_j, \mathfrak{b}_j$, turning P_j into Q_j , so T is injective with image the set where the least $j \geq j_0$ with $v \in P_j \cup Q_j$ has $v \in Q_j$; T is a partial transformation of $\mathcal{R}$, being the map $\mathfrak{b}_j \mathfrak{a}_j^{-1}$ on each piece $D_j = \{j(v) = j\}$, and by (2.1), $\rho(y, Ty) = \beta \log(\mathbf{N}\mathfrak{a}_j/\mathbf{N}\mathfrak{b}_j)$ lies in $(u - \varepsilon, u + \varepsilon)$ once j_0 is large. Since the D_j are disjoint and $\sum_j (p_j + q_j) = \infty$,

$$\nu(D) = \sum_{j \geq j_0} p_j \prod_{j_0 \leq i < j} (1 - p_i - q_i) \geq \theta \sum_{j \geq j_0} (p_j + q_j) \prod_{j_0 \leq i < j} (1 - p_i - q_i) = \theta.$$

Now let $\nu(B) > 0$ and $0 < \eta < 1$. The sets measurable with respect to finitely many coordinates of V together with the G_S -coordinate generate the σ -algebra of Y , so there are a finite $F \subset S$ and $A \in \sigma(v_{\mathfrak{p}} : \mathfrak{p} \in F) \otimes \mathcal{B}(G_S)$ with $\nu(A \triangle B) < \eta \nu(B)$. Translation in the second coordinate preserves ν and is continuous in $L^1(\nu)$, so $\nu(A \triangle (1 \times g)^{-1}A) \rightarrow 0$ as $g \rightarrow 1$. Choose j_0 so large that $\mathfrak{a}_j, \mathfrak{b}_j \notin F$, $|\rho - u| < \varepsilon$ on D_j , and $\nu(A \triangle (1 \times g_j)^{-1}A) < \eta \nu(B)$ for all $j \geq j_0$. For $(v, h) \in D_j$ the point $T(v, h)$ agrees with $(v, g_j h)$ on the coordinates in F , hence $D_j \cap A \cap (1 \times g_j)^{-1}A \subseteq A \cap T^{-1}A$; the set D_j depends only on the coordinates $\mathfrak{a}_i, \mathfrak{b}_i$, $i \geq j_0$, so it is independent of $A \cap (1 \times g_j)^{-1}A$, and

$$\nu(A \cap T^{-1}A) \geq \sum_{j \geq j_0} \nu(D_j) \nu(A \cap (1 \times g_j)^{-1}A) \geq \nu(D) (\nu(A) - \eta \nu(B)) \geq \theta(1 - 2\eta) \nu(B).$$

Finally $d(\nu \circ T^{-1})/d\nu = e^{-\rho \circ T^{-1}} \leq e^{|u|+\varepsilon}$ on $T(D)$, so

$$\nu(B \cap T^{-1}B) \geq \nu(A \cap T^{-1}A) - \nu(A \triangle B) - \nu(T^{-1}(A \triangle B)) \geq (\theta(1 - 2\eta) - \eta - e^{|u|+\varepsilon}\eta) \nu(B) > 0$$

for η small. As B and ε were arbitrary, $u \in E$. $\square$

The lemma is the asymptotic-ratio-set argument of Araki and Woods [1] with the G_S -coordinate carried along; the hypothesis $g_j \rightarrow 1$ is what makes that coordinate harmless. The restriction to $\Xi_\beta = \{1\}$ is one of convenience: it is the case needed below, and it is the case in which ν has the product form used in the last display.

3. THE CLASSICAL SYSTEMS

Proposition 3.1 (The slices for $S = P_K$). *Let $S = P_K$ and $0 < \beta \leq 1$. Then $\Xi_\beta^{\text{ext}}(K, S) = \{(1, 0)\}$; in particular $\Sigma_\beta = \{0\}$.*

Proof. For (χ, s) put $\tau = \beta s$; since $|1 - z|^2 = 2 - 2 \operatorname{Re} z$ for $|z| = 1$,

$$(3.1) \quad \sum_{\mathbf{N}\mathfrak{p} \leq x} \mathbf{N}\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}}) \mathbf{N}\mathfrak{p}^{i\tau}|^2 = 2 \sum_{\mathbf{N}\mathfrak{p} \leq x} \mathbf{N}\mathfrak{p}^{-\beta} - 2 \operatorname{Re} \sum_{\mathbf{N}\mathfrak{p} \leq x} \chi(\sigma_{\mathfrak{p}}) \mathbf{N}\mathfrak{p}^{-\beta+i\tau},$$

and it suffices to show that for $(\chi, \tau) \neq (1, 0)$ the second sum is o of the first. A character of $G_{P_K} = \operatorname{Gal}(K^{ab}/K)$ is a ray class character, and by [13, Lem. 3.8] $\chi(\sigma_{\mathfrak{p}}) = \overline{\chi(\operatorname{Frob}_{\mathfrak{p}})}$ off the conductor. For $\beta = 1$ the series $\sum_{\mathfrak{p}} \chi(\sigma_{\mathfrak{p}}) \mathbf{N}\mathfrak{p}^{-1+i\tau}$ differs from $\log L(1 - i\tau, \bar{\chi})$ by an absolutely convergent sum over prime powers, and $L(1 - i\tau, \bar{\chi})$ is finite — the only pole of a Hecke L -function on $\operatorname{Re} = 1$ being the simple pole of ζ_K at $s = 1$, excluded — and nonzero by the theorem of Hecke and Landau [6, 8]. So the second sum is $O(1)$ while $\sum_{\mathbf{N}\mathfrak{p} \leq x} \mathbf{N}\mathfrak{p}^{-1} \rightarrow \infty$. For $0 < \beta < 1$, partial summation from the prime ideal theorem in ray classes in the standard zero-free region [8] gives $O_A(x^{1-\beta} \log^{-A} x)$ for the second sum against $\sum_{\mathbf{N}\mathfrak{p} \leq x} \mathbf{N}\mathfrak{p}^{-\beta} \sim x^{1-\beta}/((1-\beta) \log x)$ for the first. $\square$

Theorem 3.2 (The classical systems are of type III₁). *For $S = P_K$ and $0 < \beta \leq 1$, M is of type III₁.*

For $K = \mathbb{Q}$ this is due to Bost and Connes [3], and in general to Laca, Larsen and Neshveyev [9]; we include a proof because it is the model for the computation of §4.

Proof. $\Xi_\beta = \{1\}$ by [13, Cor. 3.10], so Lemma 2.4 applies, and we must produce, for each $u \in \mathbb{R}$, pairs $(\mathfrak{a}_j, \mathfrak{b}_j)$ with $\beta \log(\mathbf{N}\mathfrak{a}_j/\mathbf{N}\mathfrak{b}_j) \rightarrow u$, $\sigma_{\mathfrak{b}_j}\sigma_{\mathfrak{a}_j}^{-1} \rightarrow 1$ in $\text{Gal}(K^{ab}/K)$ and $\sum_j \mathbf{N}\mathfrak{a}_j^{-\beta} = \infty$. Fix moduli $\mathfrak{m}_1 \mid \mathfrak{m}_2 \mid \cdots$ whose ray class fields exhaust K^{ab} [11, Ch. VI]; for primes $\mathfrak{a}, \mathfrak{b}$ in the same ray class modulo $\mathfrak{m}_k$ the element $\sigma_{\mathfrak{b}}\sigma_{\mathfrak{a}}^{-1} = \text{Frob}_{\mathfrak{a}}\text{Frob}_{\mathfrak{b}}^{-1}$ is trivial on the ray class field of $\mathfrak{m}_k$, so pairs chosen in a common class modulo $\mathfrak{m}_k$ with $k \rightarrow \infty$ satisfy $\sigma_{\mathfrak{b}_j}\sigma_{\mathfrak{a}_j}^{-1} \rightarrow 1$. By the prime ideal theorem in ray classes [8], for fixed $\mathfrak{m}$ and each class $\mathfrak{c}$ the number of primes $\mathfrak{p} \in \mathfrak{c}$ with $\mathbf{N}\mathfrak{p} \in (y, y(1+\delta))$ is $\asymp_{\mathfrak{m},\delta} y/\log y$ for $y \geq y_0(\mathfrak{m}, \delta)$. Let $r = e^{u/\beta}$. Use only every N_0 -th n , with N_0 so large that the ranges $[2^n, 2^{n+1}) \cup r^{-1}[2^n, 2^{n+1})$ of the blocks used are pairwise disjoint. Divide $[2^n, 2^{n+1})$ into intervals $I = (y, y(1+\delta_n)]$ with $\delta_n \downarrow 0$ slowly, and for each I and each class $\mathfrak{c}$ modulo $\mathfrak{m}_{k_n}$, with $k_n \rightarrow \infty$ slowly enough that $2^n \geq y_0(\mathfrak{m}_{k_n}, \delta_n)$, pair off primes of $\mathfrak{c}$ with norm in I against primes of $\mathfrak{c}$ with norm in $r^{-1}I$, injectively and disjointly from earlier pairs; the pairs of different blocks are then disjoint. Each pair has $|\beta \log(\mathbf{N}\mathfrak{a}/\mathbf{N}\mathfrak{b}) - u| \leq 2\beta\delta_n$, and the pairs with $\mathbf{N}\mathfrak{a} \in [2^n, 2^{n+1})$ number $\gg 2^n/n$ (for large n , a positive proportion of the primes in $[2^n, 2^{n+1})$ is used), so they contribute $\gg 2^{n(1-\beta)}/n \geq 1/n$ to $\sum_j \mathbf{N}\mathfrak{a}_j^{-\beta}$, which diverges along the blocks used. Lemma 2.4 gives $u \in E$ for every u , so $E = \mathbb{R}$ and M is of type III_1 by (2.2). $\square$

Remark 3.3 (Two slices of one analytic input). At $\beta = 1$ and $S = P_K$ the $s = 0$ slice of Ξ_β^{ext} is the uniqueness half of the classical theorem, equivalent by [13, Rem. 3.11] to $L(1, \chi) \neq 0$; the slices $s \neq 0$ are governed, by Proposition 3.1, by $L(1 - i\tau, \chi) \neq 0$, and their triviality is the necessary condition for type III_1 of Proposition 2.2(1). The classification of the phase and the type of the factor are thus the two coordinate slices of one obstruction group, and for the full prime set both are governed by the non-vanishing of Hecke L -functions on the line $\text{Re} = 1$. That the $s \neq 0$ slices are trivial is, by Remark 2.3, the statement that the flow of weights has no nonzero eigenvalue; that M is III_1 , i.e. that the flow is trivial, needed the additional input of Lemma 2.4.

4. TWO INCOMMENSURABLE NORMS DO NOT SUFFICE

Lemma 4.1 (No exact commensurability). *Let S be infinite. Then the numbers $\log \mathbf{N}\mathfrak{p}$, $\mathfrak{p} \in S$, are never pairwise commensurable.*

Proof. $\mathbf{N}\mathfrak{p} = p^{f_p}$ with $f_p \leq [K : \mathbb{Q}]$, and only finitely many primes of K lie above a given p , so an infinite S meets infinitely many rational primes; and $\log p_1/\log p_2 \notin \mathbb{Q}$ for distinct rational primes since $p_1^m = p_2^n$ forces $m = n = 0$. $\square$

So exact commensurability, which would put all the values of the Radon–Nikodym cocycle (2.1) in a lattice, never occurs for infinite S , and any failure of type III_1 must come from *approximate* commensurability. It does.

Theorem 4.2 (Exact determination of a III_{e-1} system). *Let $K = \mathbb{Q}$, $\beta = 1$, $\|\cdot\|$ the distance to the nearest integer, and*

$$(4.1) \quad S = \{3\} \cup \{\ell \text{ prime} : \ell \geq 16, \|\log \ell\| \leq \varepsilon(\ell)\}, \quad \varepsilon(\ell) = (\log \log \ell)^{-1/2}.$$

Then $\zeta_S(1) = \infty$, S contains primes with irrationally related logarithms, and

$$\Xi_1^{\text{ext}}(\mathbb{Q}, S) = \{1\} \times 2\pi\mathbb{Z}, \quad \Sigma_1 = 2\pi\mathbb{Z}.$$

Hence $\Xi_1 = \{1\}$ — the KMS_1 state is unique — while M is of pure type III_{e-1} ; both III_1 and III_0 are excluded.

The proof occupies the rest of this section. The mechanism is a band decomposition: because (4.1) localizes $\log \ell$ near an *integer*, the rounded logarithm is constant on each band and the exponential sums reduce to geometric series.

Lemma 4.3 (Bands). *Put $\varepsilon_n = (\log n)^{-1/2}$ and $n_0 = 55$, so $\varepsilon_n < \frac{1}{2}$ for $n > n_0$. For $n > n_0$ let $B_n = \{\ell \in S : |\log \ell - n| \leq \varepsilon(\ell)\}$ and $S_0 = \{\ell \in S : \ell \leq e^{n_0+1}\}$. Then:*

- (1) $S = S_0 \sqcup \bigsqcup_{n > n_0} B_n$ and $\sum_{\ell \in S_0} \ell^{-1} = O(1)$;
- (2) for $\ell \in B_n$, $\log \ell = n + \delta_\ell$ with $|\delta_\ell| \leq \varepsilon_n(1 + O(n^{-1}))$, so $\text{round}(\log \ell) = n$ is constant on B_n ;
- (3) $w_n := \sum_{\ell \in B_n} \ell^{-1} = \frac{2\varepsilon_n}{n}(1 + o(1)) = \frac{2}{n\sqrt{\log n}}(1 + o(1))$, eventually decreasing to 0 up to an absolutely summable perturbation;
- (4) $W(x) := \sum_{\ell \in S, \ell \leq x} \ell^{-1} = 4\sqrt{\log \log x} + O(1) \rightarrow \infty$.

Proof. (1) For $n > n_0$ the bands lie in the pairwise disjoint intervals $[e^{n-1/2}, e^{n+1/2}]$; for $\ell \leq e^{n_0+1}$ one has $\varepsilon(\ell) > 1/2$ and the condition may hold for every ℓ , so S_0 can be all primes up to e^{n_0+1} , of Mertens sum $O(1)$. (2) $|\log \ell - n| < \frac{1}{2}$ gives the rounding, and $\log \log \ell = \log n + O(\varepsilon_n/n)$ gives $\varepsilon(\ell) = \varepsilon_n(1 + O(n^{-1}))$. (3) By (2), B_n is the set of primes in $[e^{n-\varepsilon_n}, e^{n+\varepsilon_n}]$; Mertens with error $O(e^{-c\sqrt{\log y}})$ gives $w_n = \log \frac{n+\varepsilon_n}{n-\varepsilon_n} + O(e^{-c\sqrt{n}})$, which is the stated asymptotic, with main term $2n^{-1}(\log n)^{-1/2}$ strictly decreasing and both errors absolutely summable. (4) $\sum_{n \leq N} 2n^{-1}(\log n)^{-1/2} = 4\sqrt{\log N} + O(1)$ by comparison with the integral; put $N = \lfloor \log x \rfloor$. $\square$

Lemma 4.4 (Linearization). *Let χ factor through $(\mathbb{Z}/M\mathbb{Z})^\times$ with Dirichlet character $\tilde{\chi}$, so that $\chi(\sigma_\ell) = \tilde{\chi}(\ell)$ for $\ell \nmid M$ by [13, Lem. 2.10(1)], and let F be the set of primes dividing M . For $s \in \mathbb{R}$ and $N = \lfloor \log x \rfloor$,*

$$(4.2) \quad \sum_{\ell \in S, \ell \leq x} \frac{|1 - \chi(\sigma_\ell) \ell^{is}|^2}{\ell} = 2W(x) - 2\text{Re } \Lambda(x) + O_M(1), \quad \Lambda(x) = \sum_{n \leq N} e^{isn} c_n + O\left(|s| \sum_{n \leq N} \varepsilon_n w_n\right),$$

where $c_n = \sum_{\ell \in B_n \setminus F} \tilde{\chi}(\ell) \ell^{-1}$. Moreover

$$(4.3) \quad |s| \sum_{n \leq N} \varepsilon_n w_n = 2|s| \log \log N + O(|s|) = o(W(x)).$$

Proof. $|1 - z|^2 = 2 - 2\text{Re } z$ for $|z| = 1$ gives the first identity. For $\ell \in B_n$, $\ell^{is} = e^{is(n+\delta_\ell)} = e^{isn} + O(|s|\varepsilon_n)$ by $|e^{i\theta} - 1| \leq |\theta|$, and summing against ℓ^{-1} over B_n gives the second. For (4.3), $\varepsilon_n w_n = 2\varepsilon_n^2/n(1 + o(1)) = 2/(n \log n)(1 + o(1))$ and $\sum_{n \leq N} (n \log n)^{-1} = \log \log N + O(1)$; since $W(x) = 4\sqrt{\log N} + O(1)$ and $\log \log N = o(\sqrt{\log N})$, the claim follows. In the variable x this reads

$$(4.4) \quad |s| \sum_{n \leq N} \varepsilon_n w_n \asymp 2|s| \log \log \log x = o\left(\sqrt{\log \log x}\right) = o(W(x)),$$

the comparison being with $\sqrt{\log \log x}$ and not with $\log \log x$, since $W(x) = 4\sqrt{\log \log x} + O(1)$ by Lemma 4.3(4). $\square$

Remark 4.5 (The one delicate point). Estimate (4.3) is where care is needed: the error from linearizing ℓ^{is} across a band *diverges*, at rate $\log \log \log x$, and is absorbed only because $W(x)$ diverges faster, at rate $\sqrt{\log \log x}$. The exponent $-\frac{1}{2}$ in $\varepsilon(\ell) = (\log \log \ell)^{-1/2}$ is exactly what makes the convergence in Proposition 4.8 and this absorption hold simultaneously: an exponent $-\frac{1}{3}$ would break the former and -1 the latter.

Lemma 4.6 (Cancellation in bands). *If $\tilde{\chi}$ is non-principal modulo M then $|c_n| \ll_M e^{-c\sqrt{n}}$, so $\sum_n |c_n| < \infty$.*

Proof. Siegel–Walfisz gives $\sum_{p \leq y, p \equiv a(M)} p^{-1} = \varphi(M)^{-1} \log \log y + C(M, a) + O_M(e^{-c\sqrt{\log y}})$. Applying this at the two endpoints of B_n , whose logarithms are $n + O(1)$, the constants $C(M, a)$ cancel

and $\sum_{\ell \in B_n, \ell \equiv a} \ell^{-1} = w_n / \varphi(M) + O_M(e^{-c\sqrt{n}})$. Hence $c_n = \frac{w_n}{\varphi(M)} \sum_a^* \tilde{\chi}(a) + O_M(\varphi(M)e^{-c\sqrt{n}})$, and the character sum vanishes. $\square$

Proposition 4.7 (Non-trivial characters). *If $\chi \neq 1$ then $\sum_{\ell \in S} \ell^{-1} |1 - \chi(\sigma_\ell) \ell^{is}|^2 = \infty$ for every $s \in \mathbb{R}$. In particular $\Xi_1^{\text{ext}} \subseteq \{1\} \times \mathbb{R}$ and, taking $s = 0$, $\Xi_1 = \{1\}$.*

Proof. First, $\sigma(I_S)$ is dense in G_S : S contains all primes up to e^{n_0+1} by Lemma 4.3(1) and, by Lemma 4.6 applied to the principal character, meets every reduced class modulo M in every large band; so the classes of $S \setminus F$ generate $(\mathbb{Z}/M\mathbb{Z})^\times$ and $\chi \mapsto \tilde{\chi}$ is injective, whence $\tilde{\chi}$ is non-principal. Now combine (4.2), Lemma 4.6 and (4.3): $\Lambda(x) = O(\sum_n |c_n|) + o(W(x)) = o(W(x))$, so the left side of (4.2) is $2W(x)(1 + o(1)) \rightarrow \infty$. $\square$

Proposition 4.8 (The trivial character). *$\sum_{\ell \in S} \ell^{-1} |1 - \ell^{is}|^2$ is finite for $s \in 2\pi\mathbb{Z}$ and infinite for $s \notin 2\pi\mathbb{Z}$.*

Proof. Let $s = 2\pi m$. For $\ell \in B_n$, $\log \ell = n + \delta_\ell$ with $n \in \mathbb{Z}$, so $\ell^{is} = e^{2\pi i m \delta_\ell}$ and $|1 - \ell^{is}|^2 = 4 \sin^2(\pi m \delta_\ell) \leq 16\pi^2 m^2 \varepsilon_n^2$; hence the sum is $\ll m^2 \sum_n \varepsilon_n^2 w_n \ll m^2 \sum_n n^{-1} (\log n)^{-3/2} < \infty$.

Let $s \notin 2\pi\mathbb{Z}$. Here $\tilde{\chi}$ is principal, so $c_n = w_n + O_M(1) \mathbf{1}_{n \leq n_F}$. Since w_n is monotone to 0 up to an absolutely summable perturbation (Lemma 4.3(3)) and $|\sum_{n \leq N} e^{isn}| \leq |\sin(s/2)|^{-1} < \infty$, Dirichlet's test gives convergence of $\sum_n e^{isn} w_n$. With (4.3), $\Lambda(x) = O_s(1) + o(W(x)) = o(W(x))$ and the sum diverges as before. $\square$

Lemma 4.9 (Matched pairs in consecutive bands). *There are pairs (a_j, b_j) of primes of S with $a_j \in B_{n_j+1}$, $b_j \in B_{n_j}$, all the a_j, b_j distinct, such that $\log(a_j/b_j) \rightarrow 1$, $\sigma_{b_j} \sigma_{a_j}^{-1} \rightarrow 1$ in $G_S = \widehat{\mathbb{Z}}_S^\times$, and $\sum_j a_j^{-1} = \infty$.*

Proof. For $\ell \in B_n$, $\log \ell = n + \delta_\ell$ with $|\delta_\ell| \leq 2\varepsilon_n$, so any $a \in B_{n+1}$, $b \in B_n$ satisfy $|\log(a/b) - 1| \leq 4\varepsilon_n \rightarrow 0$. Put $k_n = \lfloor \log \log n \rfloor$ and $M_n = \prod_{\ell' \leq k_n} \ell'^{k_n}$, so that $\log M_n \leq 2k_n^2 = o(\log n)$. By Siegel–Walfisz, uniformly for moduli $q \leq (\log y)^A$ one has $\pi(y; q, a) = \text{li}(y)/\varphi(q) + O_A(ye^{-c\sqrt{\log y}})$; applied at the endpoints $e^{n \pm \varepsilon_n^\pm}$ of B_n with $q = M_n \leq n$, and since $|B_n| \asymp \varepsilon_n e^n / n$ while the error is $O(e^{n-c\sqrt{n}})$, every reduced class a modulo M_n contains at least $\frac{1}{2}|B_n|/\varphi(M_n)$ primes of B_n and at least as many primes of B_{n+1} , for all large n . Pairing, injectively within each class, primes of B_{n+1} with primes of B_n in the same class gives at least $\frac{1}{2}|B_n|$ pairs $(a, b) \in B_{n+1} \times B_n$ with $a \equiv b \pmod{M_n}$; pairs from different n are disjoint because the bands are.

For a fixed prime $\ell' \in S$ and $k \geq 1$ we have $\ell'^k \mid M_n$ for all large n , and then the ℓ' -component of $\sigma_b \sigma_a^{-1}$, which is b/a for $\ell' \notin \{a, b\}$ [13, Lem. 2.10(1)], is $\equiv 1 \pmod{\ell'^k}$; so every coordinate of $\sigma_{b_j} \sigma_{a_j}^{-1}$ tends to 1, which is convergence to 1 in the product topology of $\widehat{\mathbb{Z}}_S^\times$. Finally, the pairs from band n contribute

$$\sum_a a^{-1} \geq \frac{1}{2} |B_n| e^{-(n+2)} \geq \frac{1}{2e^3} w_n, \quad \text{since } w_n = \sum_{\ell \in B_n} \ell^{-1} \leq |B_n| e^{-(n-1)},$$

and $\sum_n w_n = \infty$ by Lemma 4.3(4). $\square$

Proof of Theorem 4.2. $\zeta_S(1) = \infty$ is Lemma 4.3(4). S is infinite, so Lemma 4.1 supplies two primes with irrationally related logarithms. Propositions 4.7 and 4.8 give $\Xi_1^{\text{ext}} = \{1\} \times 2\pi\mathbb{Z}$, hence $\Xi_1 = \{1\}$ and $\Sigma_1 = 2\pi\mathbb{Z}$. Proposition 2.2(3) with $c = 2\pi$ gives $E \subseteq \mathbb{Z}$: M is of type $\text{III}_{e^{-k}}$ for some $k \geq 1$ or of type III_0 , and in particular not of type III_1 . Since $\Xi_1 = \{1\}$, Lemma 2.4 applies, and Lemma 4.9 supplies pairs with $\beta \log(a_j/b_j) \rightarrow 1$ ($\beta = 1$), $\sigma_{b_j} \sigma_{a_j}^{-1} \rightarrow 1$ and $\sum a_j^{-1} = \infty$; hence $1 \in E$, so $E = \mathbb{Z}$, $S(M) = \{0\} \cup e^\mathbb{Z}$ by (2.2), and M is of type $\text{III}_{e^{-1}}$. $\square$

Remark 4.10 (What the proof does not use). Only Mertens, Siegel–Walfisz at a *fixed* modulus, Abel summation and Dirichlet's test are needed. Because $\text{round}(\log \ell)$ is constant on each band,

the exponential sum in (4.2) is over the *integers* n , not over primes: its partial sums are geometric and bounded by $|\sin(s/2)|^{-1}$, so Dirichlet's test handles all $s \notin 2\pi\mathbb{Z}$ at once, with no distinction between rational and irrational $s/2\pi$ and no discrepancy estimate. No Erdős–Turán inequality, no Vinogradov-type bound for $\sum_\ell \ell^{-1+is}$, and no uniformity in the modulus is required. The bands are not short intervals in the difficult sense: B_n has multiplicative length $e^{2\varepsilon_n}$ with $\varepsilon_n \asymp (\log \log x)^{-1/2}$, so $|B_n| \gg x/(\log x)(\log \log x)^{1/2}$, far above any threshold at which primes in progressions become hard.

Remark 4.11 (Numerical check). The asymptotic regime of (4.1) begins at $\varepsilon_n < \frac{1}{2}$, i.e. $\ell > e^{e^4} \approx 10^{23}$, and is beyond direct computation. The mechanics of Lemmas 4.3(3) and 4.6 were instead verified with a fixed band half-width $\varepsilon = 0.05$, for which the same computations apply verbatim: over primes up to $6 \cdot 10^7$ and bands $8 \leq n \leq 17$, the ratio $w_n / \log \frac{n+\varepsilon}{n-\varepsilon}$ lies in $[0.998, 1.003]$ for $n \geq 13$, and for the non-principal character modulo 4 the ratio $|c_n|/w_n$ falls from $8.6 \cdot 10^{-2}$ at $n = 9$ to $2.1 \cdot 10^{-4}$ at $n = 17$, consistent with Lemma 4.6.

Corollary 4.12 (Independence of the two classifications). *Uniqueness of the KMS_β state and type III_1 are logically independent: Theorem 4.2 has the first and fails the second, while Theorem 3.2 together with [13, Cor. 3.10] has both. Equivalently, Ξ_β^{ext} can be trivial on the axis $s = 0$ and nontrivial off it.*

5. THE SPECTRUM OF TRANSITIONS

We now turn to the horizontal direction: which chains $\{\Xi_\beta\}_{\beta>0}$ occur. By [13, Lem. 3.5] the chain is non-decreasing, and by [13, Prop. 4.16] any finite chain is realizable. We realize arbitrary countable chains.

Theorem 5.1 (Realization). *Let $(\beta_n)_{n \geq 1} \subset (0, 1)$ be any sequence. There is $S = Q \sqcup P_0 \sqcup \bigsqcup_n R_n \subset P_{\mathbb{Q}}$ with*

- (1) $\beta_c(S) = 1$ and $\zeta_S(\beta) = \infty$ for $0 < \beta \leq 1$;
- (2) $\sigma(I_S)$ dense in G_S ;
- (3) with q_n the n -th element of Q and χ_n the quadratic character at that component,

$$\Xi_\beta(\mathbb{Q}, S) = \langle \chi_n : \beta_n < \beta \rangle \cong \bigoplus_{n: \beta_n < \beta} \mathbb{Z}/2\mathbb{Z} \quad (0 < \beta \leq 1),$$

and $\Xi_\beta = \widehat{G_S}$ for $\beta > 1$. In particular the jump set is $\{\beta_n\} \cup \{1\}$.

The construction proceeds in blocks; the block scheme is what makes infinitely many congruence conditions manageable.

Lemma 5.2 (Independence of the quadratic conditions). *Let $q_1 < \dots < q_j$ be distinct odd primes, $q^* = (-1)^{(q-1)/2}q$, $M \geq 1$, and $E = \mathbb{Q}(\zeta_M)(\sqrt{q_1^*}, \dots, \sqrt{q_j^*})$. Then $[E : \mathbb{Q}] = \varphi(M) \cdot 2^{\#\{i \leq j : q_i \nmid M\}}$, and for any $a \in (\mathbb{Z}/M\mathbb{Z})^\times$ and signs ϵ_i compatible with a at the q_i dividing M , the primes ℓ with $\ell \equiv a \pmod{M}$ and $\left(\frac{\ell}{q_i}\right) = \epsilon_i$ have natural density $[E : \mathbb{Q}]^{-1} > 0$.*

Proof. The q_i^* are multiplicatively independent modulo squares, so $[\mathbb{Q}(\sqrt{q_1^*}, \dots, \sqrt{q_j^*}) : \mathbb{Q}] = 2^j$ by Kummer theory. $\mathbb{Q}(\sqrt{q_i^*})$ is the unique quadratic subfield of $\mathbb{Q}(\zeta_{q_i})$, so it lies in $\mathbb{Q}(\zeta_M)$ iff $q_i \mid M$; the degree formula follows. Quadratic reciprocity turns the sign conditions into splitting conditions, and Chebotarev applies. $\square$

Definition 5.3 (Block scheme). Choose $z_1 < z_2 < \dots$ recursively, z_{j+1} as large as needed in terms of z_j and j , and put $B_j = [z_j, z_{j+1})$. Let q_j be a prime in B_j with $\left(\frac{q_j}{q_i}\right) = +1$ for all $i < j$

(such primes have positive density by Lemma 5.2, so they exist in B_j once z_{j+1} is large), and $Q = \{q_j\}$. For $j \geq 1$ set

$$C_{0,j} = \{\ell : \left(\frac{\ell}{q_i}\right) = +1 \ \forall i \leq j\}, \quad C_{n,j} = \{\ell : \left(\frac{\ell}{q_n}\right) = -1, \left(\frac{\ell}{q_i}\right) = +1 \ \forall i \leq j, i \neq n\},$$

each of density 2^{-j} by Lemma 5.2, and

$$P_0 = \bigsqcup_{j \geq 1} (C_{0,j} \cap B_j) \setminus Q, \quad R_n = \bigsqcup_{j \geq n} \rho_{n,j}, \quad \rho_{n,j} \subseteq C_{n,j} \cap B_j \text{ finite.}$$

The essential property is that for $\ell \in S \cap [z_j, \infty)$ the symbols $\left(\frac{\ell}{q_i}\right)$, $i \leq j$, are determined by which block-class ℓ lies in.

Lemma 5.4 (The pieces). *The z_j and $\rho_{n,j}$ may be chosen so that:*

- (1) $\sum_{\ell \in P_0 \cap B_j} \ell^{-1} \geq 1$ for every j , whence $\sum_{P_0} \ell^{-1} = \infty$ and $\beta_c(S) = 1$; and for every M and every $a \in (\mathbb{Z}/M\mathbb{Z})^\times$ with $\left(\frac{a}{q_i}\right) = 1$ whenever $q_i \mid M$, $\sum_{\ell \in P_0, \ell \equiv a(M)} \ell^{-1} = \infty$;
- (2) $|R_n \cap [1, x]| \asymp_n x^{\beta_n}$, hence by Abel summation $\sum_{\ell \in R_n} \ell^{-\beta} < \infty \iff \beta > \beta_n$, with divergence at the endpoint;
- (3) $q_n \geq z_n$ with $\log \log z_n \geq 2^n$, so $\sum_n q_n^{-\beta} < \infty$ for every $\beta > 0$ and Q is negligible in every convergence question.

Proof. (1) $C_{0,j}$ has density 2^{-j} , so $\sum_{\ell \in C_{0,j}, \ell \leq x} \ell^{-1} = 2^{-j} \log \log x + O_j(1)$; choose z_{j+1} with $2^{-j}(\log \log z_{j+1} - \log \log z_j) \geq 2$. For the refined statement, for j beyond the largest index with $q_i \mid M$ the two conditions are compatible and Lemma 5.2 gives positive density, so enlarging z_{j+1} further (finitely many extra requirements per block) gives block sums $\geq \varphi(M)^{-1}$. (2) $C_{n,j}$ has positive density, so it meets $[y, 2y]$ for $y \geq y_0(j)$; take $\rho_{n,j}$ to contain one prime of $C_{n,j}$ in each $[m^{1/\beta_n}, 2m^{1/\beta_n}]$ contained in B_j . Abel summation with $A(x) = |R_n \cap [1, x]| \asymp x^{\beta_n}$ and $A(t) = 0$ for $t < z_n$ gives

$$\sum_{\ell \in R_n, \ell \leq x} \ell^{-\beta} = A(x)x^{-\beta} + \beta \int_{z_n}^x A(t)t^{-\beta-1} dt \asymp x^{\beta_n-\beta} + \beta \int_{z_n}^x t^{\beta_n-\beta-1} dt.$$

The lower limit contributes the constant $z_n^{\beta_n-\beta}$; the integral converges as $x \rightarrow \infty$ iff $\beta > \beta_n$, equals $\beta_n \log(x/z_n) \rightarrow \infty$ at $\beta = \beta_n$, and diverges as a power for $\beta < \beta_n$. (3) $q_n \geq \exp \exp(2^n)$ makes $q_n^{-\beta} \leq \exp(-\beta \exp(2^n))$. $\square$

Proof of Theorem 5.1. (1) is Lemma 5.4(1). For (2), fix a modulus M supported on a finite $F \subset S$, let $Q_F = Q \cap \{q : q \mid M\}$ and $K_M = \{a : \left(\frac{a}{q}\right) = 1 \ \forall q \in Q_F\}$, a subgroup of index $2^{|Q_F|}$ with $(\mathbb{Z}/M\mathbb{Z})^\times / K_M \cong (\mathbb{Z}/2\mathbb{Z})^{Q_F}$. By Lemma 5.4(1) every $a \in K_M$ is realized by infinitely many $\ell \in P_0$, so $K_M \subseteq \langle \ell \bmod M \rangle$; and for $q_n \in Q_F$ any large $\ell \in R_n$ outside F maps to the basis vector e_n of the quotient. Hence the classes generate $(\mathbb{Z}/M\mathbb{Z})^\times$.

For (3), let $\chi \neq 1$ factor through $(\mathbb{Z}/M\mathbb{Z})^\times$. If $\tilde{\chi}|_{K_M} \neq 1$, pick $a \in K_M$ with $\tilde{\chi}(a) \neq 1$; by Lemma 5.4(1) $\sum_{\ell \in P_0, \ell \equiv a} \ell^{-1} = \infty$ and every such ℓ detects χ , so $\chi \notin \Xi_\beta$ for $\beta \leq 1$. If $\tilde{\chi}|_{K_M} = 1$ then $\tilde{\chi} = \prod_{n \in T} \tilde{\chi}_{q_n}$ for a unique T , and by (2) and injectivity $\chi = \chi_T := \prod_{n \in T} \chi_n$. For $\ell \in S$ with $\ell \geq z_{\max T}$ the block property gives $\chi_T(\sigma_\ell) = +1$ on P_0 and on R_m with $m \notin T$, and -1 on R_m with $m \in T$; so, Q being negligible and $[1, z_{\max T}]$ finite,

$$\sum_{\ell \in D_{\chi_T}} \ell^{-\beta} < \infty \iff \sum_{m \in T} \sum_{\ell \in R_m} \ell^{-\beta} < \infty \iff \beta > \max_{m \in T} \beta_m,$$

by Lemma 5.4(2). This is the stated formula; for $\beta > 1$, $\zeta_S(\beta) < \infty$ and [13, Lem. 3.5] gives $\Xi_\beta = \widehat{G}_S$. $\square$

Remark 5.5 (Numerical check). Three ingredients were verified computationally on a scaled-down model with $Q = \{7, 11, 13, 17\}$ and primes up to $2 \cdot 10^6$: the $2^4 = 16$ joint sign patterns of Lemma 5.2 occur with densities in $[0.0621, 0.0629]$ against the predicted $1/16 = 0.0625$; the detecting set of χ_T equals $\bigsqcup_{m \in T} R_m$ exactly, for every subset T tested; and a set thinned to counting function $\asymp x^b$ has its β -weighted sums collapsing across $\beta = b$ as Lemma 5.4(2) requires.

6. CANTOR TRANSITION LOCI, AND THE UNATTAINABLE STAIRCASE

Definition 6.1. The *jump set* is $J(S) = \{\beta_\chi : \chi \neq 1\}$, where β_χ is the convergence exponent of the detecting set $D_\chi = \{\mathfrak{p} \in S : \chi(\sigma_{\mathfrak{p}}) \neq 1\}$; the *transition locus* $\mathcal{L}(S)$ is the set of β near which $\beta \mapsto \Xi_\beta$ is not constant.

Corollary 6.2 (Dense and Cantor loci). *Taking (β_n) an enumeration of $\mathbb{Q} \cap (0, 1)$ in Theorem 5.1 gives $J(S) = (\mathbb{Q} \cap (0, 1)) \cup \{1\}$ and $\mathcal{L}(S) = [0, 1]$. Taking (β_n) dense in a Cantor set $\mathcal{C} \subset (0, 1)$ gives $\mathcal{L}(S) = \mathcal{C} \cup \{1\}$: the transition locus is exactly the prescribed Cantor set.*

Proof. By Theorem 5.1(3), Ξ_β is constant on an open interval $I \subseteq (0, 1)$ if and only if I contains no β_n . Hence a point of $(0, 1)$ fails to have a neighbourhood of constancy exactly when it is a limit of the β_n , so that

$$\mathcal{L}(S) \cap (0, 1) = \overline{J(S) \cap (0, 1)} = \overline{\{\beta_n\}},$$

identifying the non-local-constancy locus with the closure of the jump set, as in Remark 6.7. Finally $\beta = 1$ always lies in $\mathcal{L}(S)$, since Ξ_β jumps to $\widehat{G}_S$ there. $\square$

Definition 6.3. For weights $w_n > 0$ with $\sum w_n = 1$ the *normalized rank* is $\mathfrak{d}_w(\beta) = \sum_{n: \beta_n < \beta} w_n \in [0, 1]$.

Theorem 6.4 (Pure-jump staircase). *In the situation of Corollary 6.2 with locus $\mathcal{C}$, $\mathfrak{d}_w$ is non-decreasing, equals 0 below $\min \mathcal{C}$ and 1 above $\max \mathcal{C}$, is constant on every connected component of $(0, 1) \setminus \mathcal{C}$, and its set of points of increase is exactly $\mathcal{C}$.*

Proof. A component of the complement contains no β_n , so $\mathfrak{d}_w$ is constant there; every neighbourhood of a point of $\mathcal{C}$ contains some β_n by density. $\square$

Theorem 6.5 (The devil's staircase is unattainable). *For every set of primes S : $\widehat{G}_S$ is countable, hence $J(S)$ is countable; and for every choice of weights $\mathfrak{d}_w$ is a pure jump function, discontinuous at every β_n with jump $w_n > 0$, whose distributional derivative is the purely atomic measure $\sum_n w_n \delta_{\beta_n}$. Consequently there is no S and no weighting for which $\mathfrak{d}_w$ equals the Cantor function, or any continuous non-constant function.*

Proof. G_S is a compact metrizable abelian group, being a Hausdorff quotient of the second countable locally compact group $\mathbb{I}_S$ [13, Prop. 2.5]; the dual of a compact metrizable abelian group is countable. For each χ the set $\{\beta : \chi \in \Xi_\beta\}$ is an up-set, so each χ contributes a single jump and $J(S) = \{\beta_\chi\}$ is countable. In the situation of Theorem 5.1, $\chi_n \notin \Xi_{\beta_n}$ by the endpoint divergence in Lemma 5.4(2) while $\chi_n \in \Xi_\beta$ for $\beta > \beta_n$, so the jump at β_n is at least $w_n > 0$. A non-decreasing function of the form $\sum_{\beta_n < \beta} w_n$ has purely atomic distributional derivative. The Cantor function is continuous with non-atomic derivative, and a purely atomic probability measure and a non-atomic one are mutually singular, hence distinct. $\square$

Theorem 6.6 (Approximation). *Let $\mathcal{C}$ be the middle-thirds Cantor set, c the Cantor function, and S as in Corollary 6.2 with (β_n) dense in $\mathcal{C}$. For every $\varepsilon > 0$ there are weights $w_n > 0$, $\sum w_n = 1$, with $\|\mathfrak{d}_w - c\|_\infty < \varepsilon$.*

Proof. Let μ_c be the Cantor measure. Partition $\mathcal{C}$ into finitely many relatively clopen pieces of diameter $< \delta$ and place the μ_c -mass of each at a point β_{n_i} inside it, giving $\nu = \sum_{i \leq N} v_i \delta_{\beta_{n_i}}$.

Since c is continuous, weak convergence of distribution functions is uniform (Pólya's theorem, [2, Problem 14.8]), so for δ small $\|F_\nu - c\|_\infty < \varepsilon/2$. Put $w_{n_i} = (1 - \frac{\varepsilon}{4})v_i$ and distribute $\frac{\varepsilon}{4}$ among the remaining indices with arbitrary positive weights; then $|\mathfrak{d}_w - F_\nu| \leq \frac{\varepsilon}{4}F_\nu + \frac{\varepsilon}{4} \leq \frac{\varepsilon}{2}$. $\square$

Remark 6.7 (Jump set versus transition locus). The obstruction in Theorem 6.5 is not an artifact of the construction of §5: Ξ_β is a subgroup of the fixed countable group $\widehat{G_S}$, and a monotone family of subgroups of a countable group can change at only countably many parameters. What survives, and what Corollary 6.2 realizes, is the distinction between the *jump set*, necessarily countable, and the *transition locus* $\overline{J(S)}$, which is the closure of a countable set and hence an arbitrary separable closed set — in particular an arbitrary Cantor set. The Cantor spectrum of transitions is real; the continuity of the staircase is not. Theorems 6.5 and 6.6 together say the devil's staircase is a limit point of the realizable order parameters in the uniform norm, but not itself realizable.

7. ORDER PARAMETERS

Each transition brings into existence a macroscopic observable, and the family is independent.

Definition 7.1. With $E_n = \{\ell \in S : \ell < z_n\}$ finite and $E_n^- = \{\ell \in E_n : (\frac{\ell}{q_n}) = -1\}$, put

$$(7.1) \quad \Psi_n(x) = \left(\frac{u_{q_n}(x) \bmod q_n}{q_n} \right) \cdot (-1)^{\sum_{\ell \in R_n} v_\ell(x)} \cdot \prod_{\ell \in E_n^-} (-1)^{v_\ell(x)} \in \{\pm 1\}.$$

Proposition 7.2. (1) $\Psi_n(mx) = \Psi_n(x)$ for every $m \in \mathbb{N}_S$ and every x in the domain — pointwise, not merely almost everywhere;
 (2) Ψ_n is defined ν -almost everywhere iff $\beta > \beta_n$;
 (3) under ν_{sym} the family (Ψ_n) consists of independent uniform signs, and for $\beta > \beta_n$ the map $\varphi \mapsto (\varphi(\Psi_n))_n$ is a bijection from the extreme points of the KMS_β simplex onto $\{\pm 1\}_{\{n: \beta_n < \beta\}}$.

Proof. (1) It suffices to treat $m = \ell_0$ prime. Multiplication by ℓ_0 sends $v_\ell \mapsto v_\ell + \delta_{\ell, \ell_0}$ and $u_{q_n} \mapsto \ell_0 u_{q_n}$ for $\ell_0 \neq q_n$ [13, proof of Prop. 4.12], so the first factor is multiplied by $(\frac{\ell_0}{q_n})$ unless $\ell_0 = q_n$, when it is unchanged. If $\ell_0 = q_n$, no v_ℓ with $\ell \in R_n \cup E_n^-$ changes. If $\ell_0 \in E_n$ and the symbol is -1 then $\ell_0 \in E_n^-$ and v_{ℓ_0} increases by one, flipping the third factor. If $\ell_0 \geq z_n$ and $\ell_0 \notin R_n$, then either $\ell_0 \in Q$, and the symbol is $+1$ by the choice of Q in Definition 5.3, or the block property gives symbol $+1$; in both cases no relevant valuation changes. If $\ell_0 \in R_n$, the symbol is -1 and v_{ℓ_0} increases by one, flipping the second factor.

(2) $\sum_{\ell \in R_n} \pi_\beta(v_\ell \geq 1) = \sum_{\ell \in R_n} \ell^{-\beta}$ is finite iff $\beta > \beta_n$ by Lemma 5.4(2); apply the two Borel–Cantelli lemmas to the independent events $\{v_\ell \geq 1\}$.

(3) Under ν_{sym} the conditional law of u given v is Haar, so the u_{q_n} are independent and each $(\frac{u_{q_n}}{q_n})$ is a uniform sign independent of everything else; multiplying by the remaining factors preserves uniformity and independence. The bijection is Theorem 5.1(3) together with [13, Thm. 3.6], which identifies the extreme points with $\widehat{\Xi}_\beta = (\mathbb{Z}/2\mathbb{Z})^{\{n: \beta_n < \beta\}}$. $\square$

Remark 7.3 (Numerical check). Take the model $Q = \{7, 11, 13, 17\}$. Pointwise invariance was verified over $4 \cdot 10^5$ random configurations, with no failures, together with a control confirming that dropping the R_n -correction destroys it. The independence asserted in (3) was checked by computing $\mathbb{E}[\prod_{n \in T} \Psi_n]$ over all 15 nonempty subsets of four indices, the largest absolute value being 0.0037 against a three-sigma bound of 0.0047.

Remark 7.4 (Physical reading). Each Ψ_n is a tail observable: changing the valuations at any finite set of primes leaves it unchanged, because a change at $\ell \in R_n$ flips two factors of (7.1) simultaneously. It records one asymptotically conserved bit, coming into existence at the sharp

temperature β_n where the Borel–Cantelli series for R_n turns from divergent to convergent. As β increases across $\mathcal{C}$ the bits condense on a countable dense subset of a Cantor set, and $\mathfrak{d}_w(\beta)$ is the total weight of those already condensed; between two condensations, that is on each complementary interval, the memory is frozen. By Theorem 6.5 they cannot condense continuously: each bit is a discrete degree of freedom contributing a jump, and $\widehat{G}_S$ has only countably many. The information-theoretic counterpart — the relative entropy of an extremal state with respect to the symmetric one grows by $\log 2$ with each condensed bit — is taken up in a companion paper on invariants and entropy.

8. ASSESSMENT

Question	Answer	Reference
factor type	bounded above by Σ_β , below by pairs	Prop. 2.2, Lem. 2.4
$S = P_K$, $\beta \leq 1$	III_1 ; $\Sigma_\beta = \{0\}$ via $L(1 - i\tau, \chi) \neq 0$	Thm. 3.2, Prop. 3.1
two incommensurable norms	do not suffice	Thm. 4.2
exact commensurability	impossible for infinite S	Lem. 4.1
uniqueness $\Rightarrow \text{III}_1$	false	Cor. 4.12
chains $\{\Xi_\beta\}$	any $(\beta_n) \subset (0, 1)$ realizable	Thm. 5.1
transition locus	any Cantor set	Cor. 6.2
jump set	always countable	Thm. 6.5
devil’s staircase	unattainable, approximable	Thms. 6.5, 6.6

Five questions remain. First, whether Σ_β determines the type, i.e. whether the flow of weights of these factors always has pure point spectrum (Remark 2.3); if it does, Proposition 2.2 is an equivalence and Lemma 2.4 is not needed. Second, the extended group Ξ_β^{ext} has not been computed for the sparse sets of [13, §5]; in particular whether Brun-summable S gives III_λ or III_0 is open. In Theorem 4.2 we determined Σ_1 exactly, but for a general band set with $\varepsilon(\ell) = (\log \log \ell)^{-\theta}$ the dependence of Σ_1 on θ is unexamined. Fourth, Theorem 6.5 shows the realizable order parameters are pure jump functions; which monotone families of subgroups of a countable group arise as $\{\Xi_\beta\}$, beyond the direct sums realized here, we do not know. Finally, Lemma 2.1 says that Ξ_β is the kernel of $\chi \mapsto [\chi \circ \sigma]$ into the measurable cohomology group $H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T})$; that group itself is not computed here, and a computation would presumably exhibit $\widehat{G}_S/\Xi_\beta$ inside it and give the chain $\{\Xi_\beta\}$ a cohomological reading.

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